
DistilBERT and BERT for Text Classification: Performance, Efficiency and the Role of the Classifier Head

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Abstract

We present an evaluation of using knowledge distillation to train a smaller encoder DistilBERT from a base BERT as the teacher. We fine tune DistilBERT and BERT base under an identical classifier, optimiser and data budget on 4 text classification datasets: AG News, SST 2, Yelp Polarity and Yelp Review Full. DistilBERT holds 39 percent fewer parameters and reaches 99.0 percent of the accuracy of BERT, a mean gap of 0.90 points ranging from -0.04 to 2.29 across datasets, in exchange for 2.01 times the inference throughput and 45 percent less peak GPU memory in training. We perform a seven ablations over the classifier which results are: width and depth move F1 macro by at most 0.62 points, while freezing the transformer costs 5.79. Over 496 words, 73 percent of nearest neighbours are shared between the two pretrained encoders, at a linear CKA of 0.898.

1 Introduction

This report is a controlled comparison of two models on text classification, a BERT [Devlin et al., 2019] model with 110 million parameters and a twelve block forward pass per input, and DistilBERT [Sanh et al., 2019] following the teacher student design of knowledge distillation [Hinton et al., 2015], using six blocks in the student and the masked language modelling objective. Initially we expected to achieve 97 percent of the GLUE score in BERT with 40 percent fewer parameters as reported in [Wang et al., 2018].

The structure is based in one section by each of the following questions: how large the accuracy gap is on that dataset, what the smaller model buys in latency and memory, and whether any of the gap can be recovered by spending capacity on the classifier instead of the encoder. Both backbones get the same classifier, optimiser and data budget, on four datasets, plus an ablation over the classifier along the three conditions, freezing, width and depth, and a representation level comparison of the two encoders.

2 Approach

What distillation changed. DistilBERT keeps the hidden size of 768 and the vocabulary of BERT base, removes every second transformer block leaving six, and drops the token type embeddings and the pooler. The student is initialised from alternate teacher layers and trained with three losses: distillation against the softened teacher distribution, masked language modelling, and a cosine term aligning student and teacher hidden state directions [Sanh et al., 2019]. Depth is the axis that was cut; width, vocabulary and attention are untouched.

One classifier for both backbones. Both backbones are wrapped in the same module. The final hidden state of the [CLS] position, $h \in \mathbb{R}^{768}$, is passed through an MLP $\hat{y} =$

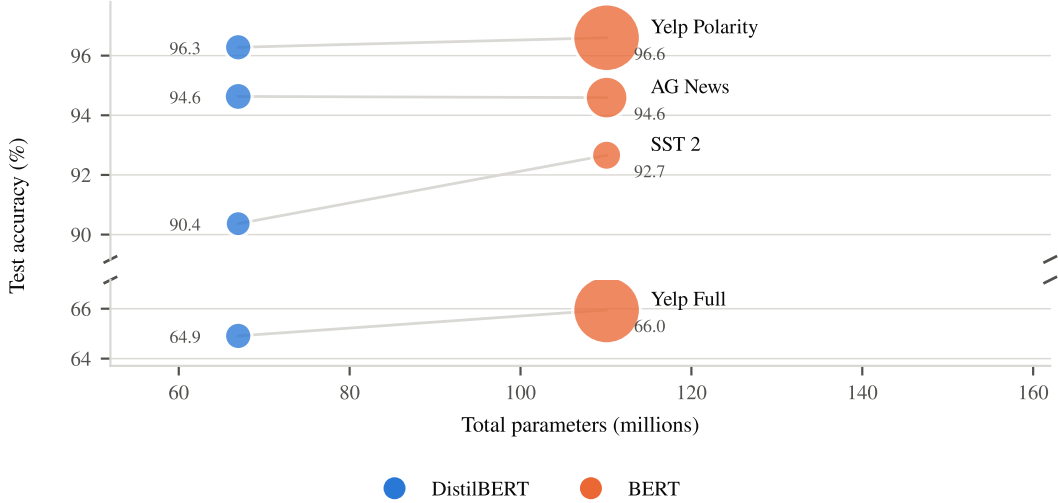


Figure 1: Parameters against accuracy. Each line joins the two backbones on one dataset and bubble area is single example GPU latency, from 2.55 to 3.99 ms. The lines are close to flat, which is this study in one picture.

$W_L \sigma(\dots \sigma(W_1 \text{drop}(h)) \dots)$ with $\sigma = \text{ReLU}$ and dropout before every linear map. We deliberately do not use the BERT pooler, which DistilBERT does not have, so both models are read the same way and no architectural advantage enters through pooling. The baseline head is one hidden layer of 768 units, reproducing the default DistilBERT design, and the same head exposes the three ablation axes: freezing covers the embeddings and the first n blocks, with n equal to all blocks giving a linear probe; width is the units per hidden layer; depth is the number of hidden layers.

What we measure. Accuracy with macro averaged precision, recall and F1 on the test split, macro averaged so the five class Yelp task is not dominated by its easiest classes; parameter count, single example latency, throughput at batch 32 and peak GPU memory. Latency is the median of 200 timed float32 passes after 20 warmup passes with device synchronisation, and memory is read from the CUDA allocator rather than the driver. Training and validation loss are recorded at twelve equally spaced iterations, the validation loss on a held out slice of train, so the test split is never touched during optimisation.

3 Experiments

Setup. AG News is four way topic classification [Zhang et al., 2015]; SST 2 is binary sentiment [Socher et al., 2013] through GLUE [Wang et al., 2018], whose labelled validation split serves as the test set as usual; Yelp Polarity and Yelp Review Full are binary and five star review sentiment [Zhang et al., 2015], each subsampled to 100,000 training and 10,000 test examples. Five percent of every training set is held out for the loss curves. Every run uses AdamW [Loshchilov and Hutter, 2019] at 2×10^{-5} on the backbone and 10^{-4} on the head, weight decay 0.01, linear decay with 6 percent warmup, batch 32, clipping at 1.0 and bfloat16 autocast, with maximum length 64 for SST 2, 128 for AG News and 256 for Yelp. Two epochs sits inside the fine tuning range recommended by Devlin et al. [2019], and the loss curves below confirm both models have converged by then. All 26 runs in this study used one NVIDIA RTX A6000 and the same seed, so latency and memory are comparable across rows.

Main comparison. Table 1 gives the four performance metrics and Figure 1 the required parameter against accuracy view. The gap averages 0.90 accuracy points in favour of BERT but is far from uniform, spanning -0.04 points on AG News to 2.29 on SST 2: BERT is clearly ahead on 3 of the 4 datasets and level on the other 1. Where a task is close to saturated the two models are indistinguishable, and depth only pays on the harder, finer grained label set.

How many epochs are needed. Both backbones are trained on SST 2 and AG News to four epochs. The best epoch varies between 2 and 4 across the 4 runs with no consistent pattern, and training

Table 1: Performance of both backbones with the identical baseline classifier. Precision, recall and F1 are macro averaged. The better value of each pair is in bold.

Dataset	Model	Accuracy	Precision	Recall	F1
AG News	DistilBERT	94.63	94.66	94.63	94.63
AG News	BERT	94.59	94.63	94.59	94.59
SST 2	DistilBERT	90.37	90.39	90.35	90.36
SST 2	BERT	92.66	92.71	92.63	92.65
Yelp Polarity	DistilBERT	96.28	96.28	96.28	96.28
Yelp Polarity	BERT	96.60	96.60	96.60	96.60
Yelp Full	DistilBERT	64.91	64.86	64.89	64.86
Yelp Full	BERT	65.95	65.81	65.93	65.86

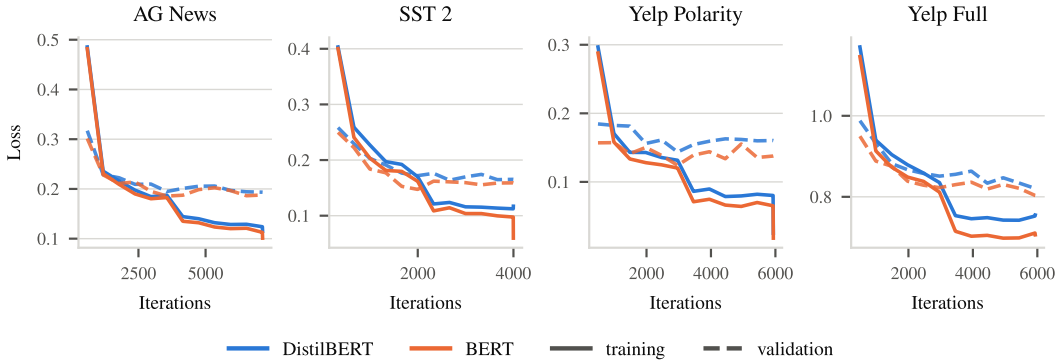


Figure 2: Iterations against training loss (solid) and validation loss (dashed). Both models flatten at the same point in the schedule; BERT settles lower.

past epoch two is worth at most 0.28 F1 macro points, inside the band we treat as noise. Therefore, two epochs seemed enough to set as the default training time for our evaluations avoiding expensive further training.

Training dynamics. We expected that DistilBERT were merely underfitting by having less parameters, additionally its training loss would sit high and still be falling at the end. However Figure 2 shows that both models flatten together, so the distance between the validation curves is a capacity gap and not an optimisation gap.

Efficiency. Halving the depth should buy a factor of two, and it does exactly that where the device is actually busy: DistilBERT sustains 1259 samples per second at batch 32 against 627, a factor of 2.01, and on CPU it needs 50.1 ms per example against 70.3. At batch one on the GPU the same models are only 1.57 times apart, 2.55 ms against 3.99 ms, because a single short sequence leaves the device idle between kernel launches and the fixed launch cost, not the twelve or six blocks, sets the clock. Peak GPU memory in training falls from 3.62 to 1.98 GiB, 45 percent less, and the four runs take 19 minutes instead of 34.

4 Analysis

The classifier head barely matters. Across the four head shape variants and the baseline, the full spread in mean F1 macro is 0.62 points (Table 2): narrowing the hidden layer to 128 units, widening it to 2048, stacking three hidden layers and removing it entirely all land inside that band. After fine tuning the [CLS] vector is close to linearly separable here, so capacity added on top of it has nothing left to extract.

Freezing is the axis that matters, and only in its upper half. Freezing the whole transformer costs 5.79 F1 macro points against the baseline, more than an order of magnitude beyond any head shape change. Freezing only the lower three blocks costs -0.05 points, inside the band that separates

Table 2: Classifier ablation on DistilBERT. Train (M) is the number of trainable parameters in millions. The best variant in each column is in bold.

Variant	Train (M)	SST 2 Acc	SST 2 F1	AG News Acc	AG News F1
A: 1x768 full	67.0	90.37	90.36	94.63	94.63
B: frozen backbone	0.6	83.37	83.37	90.07	90.04
C: 3 blocks frozen	21.9	90.94	90.93	94.17	94.17
D: 1x128	66.5	91.06	91.05	94.55	94.55
E: 1x2048	67.9	90.83	90.82	94.72	94.73
F: 3x768	68.1	90.83	90.82	94.79	94.79
G: no hidden layer	66.4	91.40	91.39	94.84	94.85

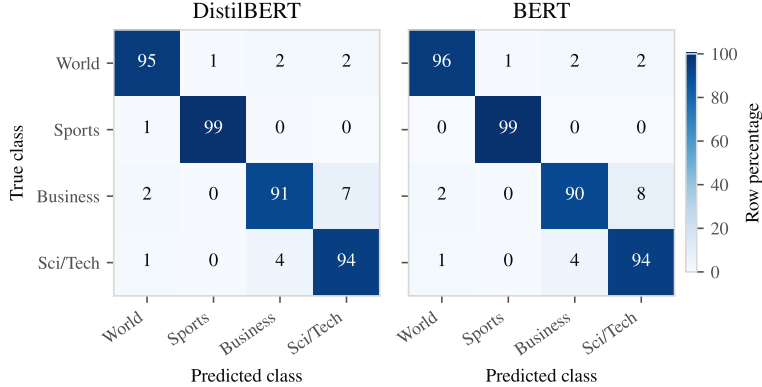


Figure 3: Row normalised confusion matrices on AG News. The two models make the same kind of mistake in the same place.

the head shapes, while cutting trainable parameters to 21.9M, 67 percent fewer, and training 1.7 times faster. By discarding these effects we conclude that task adaptation relies almost entirely in the upper half of the encoder and it is according to Kovaleva et al. [2019] that report the last two layers of BERT change most during fine tuning, Merchant et al. [2020] find that fine tuning primarily affects the top layers, and Lee et al. [2019] show that tuning only the final quarter of the layers recovers 90 percent of the quality of tuning all of them. Our runs add that the same holds for a distilled six block encoder, where there are far fewer layers to give away. The selected variant is G: no hidden layer at 93.12 mean F1 macro, the simplest head in the grid.

Best BERT version. The bests results were reported in Table 3 while Figure 3 shows where the remaining errors sit: both models concentrate them between Business and Sci/Tech, two classes that share vocabulary. The larger model resolves a few more of those borderline cases but does not fail in different ones.

Do the two encoders see language the same way? The accuracy gap says how often the models agree on a label, not whether they agree on the geometry underneath. We compare the geometries with linear centred kernel alignment, CKA [Kornblith et al., 2019]. Since it is invariant to orthogonal transformation and isotropic scaling, so a change of basis is not counted as a difference, but not to arbitrary invertible maps, this allows us to compare separately trained networks where canonical correlation analysis cannot. We embed 496 words spanning 20 topics with each pretrained encoder and compare the two 768 dimensional spaces, matching student block d to teacher block $2d$, the layer it was initialised from. Alignment is near perfect at the embeddings (0.977), still high at mid depth (0.838) and falls to 0.578 at the top, so distillation preserved the lower and middle geometry and let the specialised upper layers drift. In the layer averaged, mean centred space, 7.3 of the ten nearest neighbours BERT assigns to a word are also chosen by DistilBERT, and 72 percent of DistilBERT neighbours share their query word topic against 69 percent for BERT, so the smaller model is marginally tidier here. Fine tuning then realigns the tops: over AG News test sentences the two fine tuned encoders reach CKA 0.975. An interactive version ships with the code.

Table 3: The variant selected by the ablation (G: no hidden layer) on each backbone, with every metric this study records, measured on the same GPU. Within each dataset the better of the two values is in bold, taking lower as better for loss, size, latency, memory and time.

	SST 2		AG News	
	DistilBERT	BERT	DistilBERT	BERT
Accuracy (%)	91.40	91.97	94.84	94.72
Precision, macro (%)	91.42	92.01	94.88	94.77
Recall, macro (%)	91.38	91.95	94.84	94.72
F1, macro (%)	91.39	91.97	94.85	94.73
Test cross entropy	0.286	0.267	0.168	0.168
Total parameters (M)	66.4	109.5	66.4	109.5
Trainable parameters (M)	66.4	109.5	66.4	109.5
GPU latency, batch 1 (ms)	2.23	5.12	2.48	3.23
CPU latency, batch 1 (ms)	27.8	53.2	36.4	72.2
Throughput, batch 32 (samples/s)	2452	1218	1220	611
Peak GPU memory, training (MiB)	1313	2302	1706	3083
Peak GPU memory, inference (MiB)	404	569	539	706
Training wall clock (min)	2.6	3.9	4.3	7.8

5 Conclusion

DistilBERT keeps 99.0 percent of the accuracy of BERT base across four datasets while holding 39 percent fewer parameters, reaching 2.01 times its throughput and training in 45 percent less GPU memory, so under a latency or memory budget both have similar performance on tasks of this kind. Additionally width and depth move F1 macro by at most 0.62 points, but in the upper encoder and only there: freezing all six blocks costs 5.79 points, freezing the lower three costs -0.05 and trains 1.7 times faster, and the layerwise alignment agrees, near perfect at the embeddings and 0.578 at the top. Each run used a single seed, so differences under half a point are noise.

References

- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186, 2019. doi: 10.18653/v1/N19-1423.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. In *NeurIPS Deep Learning and Representation Learning Workshop*, 2015.
- Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of ICML*, volume 97, pages 3519–3529, 2019.
- Olga Kovaleva, Alexey Romanov, Anna Rogers, and Anna Rumshisky. Revealing the dark secrets of BERT. In *Proceedings of EMNLP-IJCNLP*, pages 4365–4374, 2019. doi: 10.18653/v1/D19-1445.
- Jaehun Lee, Raphael Tang, and Jimmy Lin. What would elsa do? freezing layers during transformer fine-tuning. *arXiv preprint arXiv:1911.03090*, 2019.
- Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.
- Amil Merchant, Elahe Rahimtoroghi, Ellie Pavlick, and Ian Tenney. What happens to BERT embeddings during fine-tuning? In *Proceedings of the EMNLP Workshop BlackboxNLP*, pages 33–44, 2020. doi: 10.18653/v1/2020.blackboxnlp-1.4.
- Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. In *5th Workshop on Energy Efficient Machine Learning and Cognitive Computing, NeurIPS*, 2019.
- Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D. Manning, Andrew Ng, and Christopher Potts. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of EMNLP*, pages 1631–1642, 2013. doi: 10.18653/v1/D13-1170.
- Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R. Bowman. GLUE: A multi-task benchmark and analysis platform for natural language understanding. In *Proceedings of the EMNLP Workshop BlackboxNLP*, pages 353–355, 2018. doi: 10.18653/v1/W18-5446.
- Xiang Zhang, Junbo Zhao, and Yann LeCun. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems*, volume 28, 2015.